

WOK AI OS – Intent Resolution & Smart Inference Specification

Objective

The recent anti-hallucination changes made WOK AI overly defensive.

Instead of intelligently understanding obvious user intent, the agents now require clarification for almost everything.

This is not the desired behavior.

WOK is an AI Operating System, not a form-filling assistant.

It should complete as much work as possible from a single natural-language request while remaining truthful and safe.

The goal of this update is to eliminate unnecessary clarification without reintroducing hallucinations.

Current Problem

The current prompts follow the rule:

Never assume.

The LLM has interpreted this as:

Never infer.

This is incorrect.

There is a major difference between **inferring user intent** and **inventing information**.

Wrong Behaviour

User:

Check my Gmail and summarize new emails from the last 24 hours.

Current reasoning:

- User did not specify a data source.
- Ask which platform they mean.
- Refuse to continue.

This is incorrect.

The user explicitly said Gmail.

WOK already has Gmail tools.

No clarification should occur.

User:

Search my Google Drive for resume.

Current behaviour:

- Which storage provider?
- Which folder?
- Which Drive?

Incorrect.

The service is already obvious.

User:

Create a Google Doc.

Current behaviour:

- Which document platform?

Incorrect.

Use docs.create.

User:

Open Chrome.

Current behaviour:

- Which application?

Incorrect.

Open Chrome.

These examples make WOK feel unintelligent.

Correct Philosophy

Replace

Never assume.

With

Never invent facts. Infer obvious intent whenever confidence is sufficiently high.

The system should understand obvious user intent exactly like a competent human assistant.

Intent Resolution Layer

Introduce a new stage before Agent A.

Pipeline:

User

↓

Intent Resolution Engine

↓

Agent A

↓

Agent B

↓

Planner

↓

QA

↓

Execution

The Intent Resolution Engine is responsible for understanding obvious intent before planning begins.

It should identify:

- Requested service
- Requested application
- Requested platform
- Requested resource
- Requested tool
- Confidence score

This resolved intent is passed to every downstream agent.

Difference Between Inference and Hallucination

Inference is allowed.

Hallucination is not.

Inference:

User:

Check Gmail.

Resolved Intent:

Service = Gmail

Confidence = 99%

Tool = gmail.search

Proceed.

Inference:

User:

Search Drive.

Resolved Intent:

Service = Google Drive

Tool = drive.search

Proceed.

Inference:

User:

Make a spreadsheet.

Resolved Intent:

Google Sheets

Tool = sheets.createTracker

Proceed.

Hallucination:

User:

Email Rahul.

System invents an email address.

Not allowed.

Hallucination:

User:

Schedule meeting tomorrow.

System invents

Time

Location

Attendees

Not allowed.

Hallucination:

User:

Open project.

System invents project name.

Not allowed.

The distinction is simple:

Infer services.

Never invent execution data.

Confidence-Based Decision Engine

Every inferred intent should receive a confidence score.

Confidence $\geq$ 95%

Proceed automatically.

No clarification.

Confidence between 70% and 95%

Proceed using the most likely interpretation.

Mention the assumption only if it affects execution.

Confidence below 70%

Ask the user.

Do not guess.

Smart Clarification Rules

Clarification should become the final option instead of the first option.

Current behaviour:

Unknown?

↓

Ask.

New behaviour:

Can the intent be inferred?

↓

Yes

↓

Continue.

↓

No

↓

Ask.

The system should exhaust every reasonable inference before interrupting the user.

Default Intent Mapping

When the user explicitly names a service or application, automatically map it.

Examples include:

Gmail

Inbox

Email

→ Gmail tools

Google Drive

Drive

Folder

→ Drive tools

Google Docs

Document

Doc

→ Docs

Spreadsheet

Excel

Google Sheet

→ Sheets

Presentation

Slides

PowerPoint

→ Slides

Calendar

Meeting

Schedule

→ Calendar

Browser

Chrome

Edge

Firefox

→ Browser/OpenApp

Terminal

CMD

PowerShell

Bash

→ Terminal

Contacts

Phonebook

→ Contacts

Reminder

Notification

Alert

→ Notifications

Memory

Remember

Preference

→ Memory

No clarification should be requested for these obvious mappings.

Smart Context Inference

The system should use:

Current conversation

Previous conversation context

Available memory

Available connected services

Known installed applications

Known authenticated accounts

Known available tools

To infer user intent whenever safe.

If WOK already knows the user is connected to Gmail, then "check my email" should immediately use Gmail.

If WOK knows only one Google account is connected, it should use that account.

If only one browser exists, use it.

Avoid unnecessary clarification when there is only one valid interpretation.

Clarification Should Only Occur When

Clarification is required only if multiple valid interpretations would result in different executions.

Examples:

Email Rahul.

Multiple Rahuls exist.

Ask.

Open project.

Multiple projects exist.

Ask.

Delete report.

Multiple reports exist.

Ask.

Schedule meeting tomorrow.

Time missing.

Ask.

These are execution-critical ambiguities.

Everything else should continue automatically.

New Agent Behaviour

Every agent should distinguish between:

Known

Highly Likely

Possible

Unknown

Impossible

Only Unknown and Impossible should stop execution.

Highly Likely should proceed.

New Global Rule

Replace

Never Assume

With

Infer obvious intent whenever the inference is strongly supported by the user's wording, conversation context, connected services, available tools, installed applications, authenticated accounts, or established defaults.

Never invent facts that directly affect execution.

Never fabricate parameters.

Never fabricate resources.

Never fabricate contacts.

Never fabricate dates.

Never fabricate times.

Never fabricate identifiers.

Never fabricate API results.

Ask for clarification only when multiple valid interpretations would change the actual execution or introduce meaningful risk.

Desired User Experience

The user should be able to type:

"Check my Gmail."

"Search my Drive."

"Create a Google Doc."

"Summarize today's emails."

"Open Chrome."

"Create a presentation."

"Find my resume."

"Show tomorrow's meetings."

"Search my downloads."

Without answering unnecessary follow-up questions.

WOK should behave like an intelligent operating system that understands obvious intent while remaining truthful, deterministic, and safe.

The objective is to maximize autonomous task completion while minimizing unnecessary clarification, without allowing hallucinations or fabricated execution details.